

Mathematics

Year 8

Student Manual



Prime Books • Mathematics Year 8 • Student Book

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Every scenario, dataset, name and worked example in this book is original to this edition. Factual and historical references are drawn from real, credited sources, which are listed in the Sources & references section at the back of this book.

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Printed for Prime School • Year 8 • Mathematics

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1.2	Multiplying and dividing integers	Apply sign rules in multi-step calculations
1.3	Square roots and cube roots	Work with roots and estimate their values
1.4	Indices	Use index notation and the laws of indices

Unit 1 also includes: a Final project, Check your progress, Evaluation, What can you do, and a Keywords round-up.

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INTRODUCTION

How this book works

Welcome to Prime Books Mathematics for Year 8. This book is written for you, the pupil, to read, argue with and write in. It is not a book to be afraid of. Every idea is introduced with a reason before it becomes a rule, and every rule is followed by the chance to use it until it feels obvious.

Mathematics at this level is about making the familiar strange and the strange familiar. You already know how to add, subtract, multiply and divide. In this first unit you will stretch those four operations across the whole number line, into the territory of the negatives, and then use them to unlock older and deeper patterns: multiples, factors, quick tests for divisibility, and the roots that undo powers.

What makes this book different

- **Every concept has a reason.** You will rarely meet a bare rule. You will meet a situation that forces the rule into being.
- **Every example is worked in full.** Nothing is left as "an exercise for the reader" in the middle of a new idea.
- **You are asked to think like a mathematician.** Several times in each unit you are invited to investigate, conjecture and justify, not just calculate.
- **Everything is real.** Names, places and stories are drawn from life in Portugal, and every fact can be checked in the Sources & references at the back.

A note to teachers and parents

This book follows the spirit and sequence of an international lower-secondary mathematics programme, interpreted afresh for Prime School. Each unit opens with diagnostic questions to surface prior knowledge, builds concepts through worked examples and structured practice, and closes with self-assessment and a summative check. The pedagogy is deliberately investigation-led: pupils are asked to notice patterns before being handed formulas, because mathematics that is discovered is mathematics that is remembered.

HOW TO USE THIS BOOK

Reading the page

Each unit follows the same rhythm, so you always know where you are. Here is what each kind of panel is for.

PANEL	WHAT IT DOES
Get started	A few quick questions to wake up what you already know. No new ideas here.
Scenario	A real situation that makes the unit's maths worth learning.
Learning outcomes	The promises this unit makes to you. Tick them off as you master them.
Warm up	A short puzzle to get your mathematical blood moving.
Do you remember?	A gentle reminder of earlier work the new idea depends on.
Learn	The main teaching. Read slowly; this is where new ideas are born.
Worked example	A problem solved in full, with the reasoning shown beside it.
Practise	Questions for you. Fluency first, then reasoning, then a stretch.
Did you know?	A true and surprising fact connected to the topic.
Think like a mathematician	An open investigation. Conjecture, test, justify.
Go further	For when you want more: a harder idea or a wider view.
Challenge yourself	The stretch. Attempt these; they are meant to be hard.
Final project	A bigger task that pulls the whole unit together.
Check your progress	A mixed set of questions across the whole unit.
Evaluation	Be honest with yourself: green, amber or red on each outcome.
What can you do?	A checklist to confirm what you can now do.
Keywords	The mathematical words this unit has taught you.

How to work

Read with a pencil in your hand. When the book asks a question, stop and try it before you read the answer. Mathematics is not a spectator sport: you learn it by doing it, getting it wrong, and seeing why.

GETTING SET UP

Before Unit 1

You do not need much for this unit: a pencil, a ruler, a rubber, and a squared exercise book. A calculator is useful for checking, but most of this unit is about doing without one, so keep it in your bag until you are asked.

Your mathematical toolkit for this unit

- A **number line** you can sketch quickly, with zero in the middle.
- Times tables up to 12×12 , fluent enough that you do not have to think.
- The four operations on positive whole numbers, secure.
- Curiosity. This is the most important tool.

A quick self-check

Try these now. If any feel hard, tell your teacher before you begin: the unit builds on them.

- 1 Write the numbers -3 , 5 , 0 , -7 , 2 in order, smallest first.
- 2 List the first four multiples of 7 .
- 3 List all the factors of 20 .
- 4 Work out 13×4 and $96 \div 8$ without a calculator.

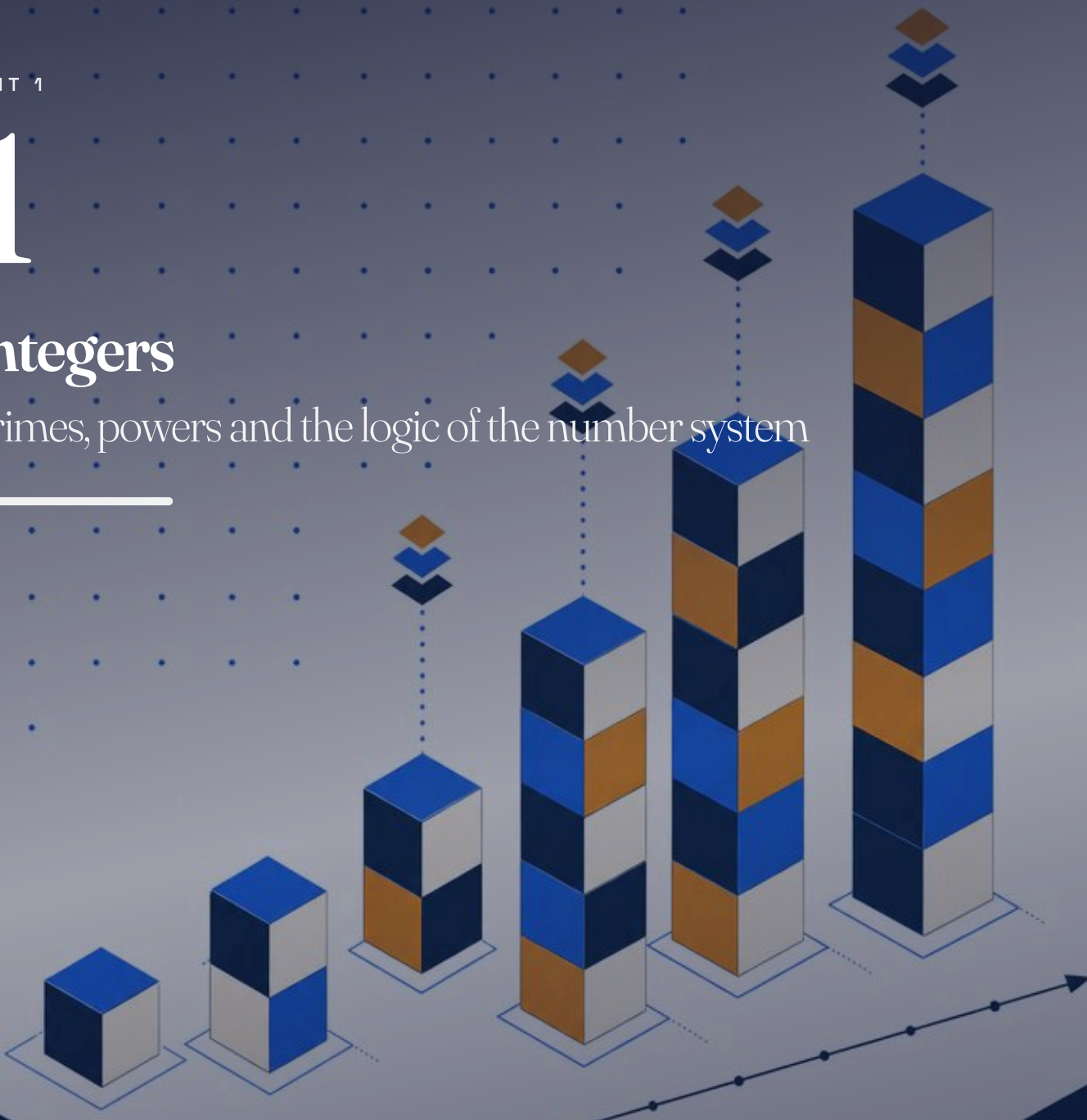
The answers, and the ideas behind them, are exactly what Unit 1 will sharpen. When you can do these confidently, you are ready.

UNIT 1

1

Integers

Primes, powers and the logic of the number system



© LEARNING OUTCOMES

- ✓ Find factors, multiples and prime numbers.
- ✓ Write a number as a product of its prime factors.
- ✓ Multiply and divide integers in multi-step calculations.
- ✓ Work with square roots, cube roots and estimates.
- ✓ Use index notation and the laws of indices.

◆ GET STARTED

- 1 List all the factors of 36.
- 2 Write down the first five multiples of 9.
- 3 Work out: a. -7×6 b. $(-48) \div (-6)$.
- 4 Write $4 \times 4 \times 4$ in index form.

◆ SCENARIO

Every year, the organisers of the Sardinha Festival in a small Portuguese town have to arrange the sardine grills in perfect rows. If they have 84 grills, they can make 7 rows of 12, or 12 rows of 7, or 6 rows of 14. The numbers that divide 84 exactly decide which arrangements work. This unit is about the hidden structure of numbers like 84: their factors, their multiples, and the prime building blocks underneath them all.

UNIT 1 • TOPIC 1.1

Factors, multiples and primes

Learn

A **factor** of a number divides it exactly. A **multiple** of a number is what you get by multiplying it by 1, 2, 3 and so on. Some numbers have exactly two factors: 1 and themselves. These are the **prime numbers**, and they are the atoms of arithmetic. The primes up to 30 are 2, 3, 5, 7, 11, 13, 17, 19, 23 and 29.

Every whole number greater than 1 can be written as a product of prime numbers in exactly one way. This is called writing it as a **product of prime factors**.

■ WORKED EXAMPLE 1.1

Write 84 as a product of prime factors.

$$\begin{aligned}84 \div 2 &= 42 \\42 \div 2 &= 21 \\21 \div 3 &= 7 \\7 \div 7 &= 1 \\ \text{so } 84 &= 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7\end{aligned}$$

Why: Dividing out each prime in turn breaks 84 down to its building blocks. The result is unique: no other collection of primes multiplies to 84.

- **PRACTISE 1.1**

- 1 List all the factors of: a. 45 b. 60.
- 2 Write down the prime numbers between 30 and 50.
- 3 Write each number as a product of prime factors:
a. 60 b. 72 c. 90.
- 4 Is 51 a prime number? Explain.
- 5 The festival organisers have 84 grills. List all the possible numbers of equal rows they can make.

- **KEYWORDS**

- **factor:** a number that divides another exactly.
- **multiple:** the result of multiplying a number by a positive integer.
- **prime number:** a number with exactly two factors, 1 and itself.
- **prime factor:** a factor that is a prime number.

UNIT 1 • TOPIC 1.2

Multiplying and dividing integers

Learn

You met the sign rules in Year 7: same signs give a positive result, different signs give a negative result. Now you will use them in longer calculations, always working inside brackets first.

- **WORKED EXAMPLE 1.2**

Work out $(-3) \times (-8) + (-48) \div 6$.

$(-3) \times (-8) = 24$	same signs: positive
$(-48) \div 6 = -8$	different signs: negative
$24 + (-8) = 16$	

Why: Each multiplication and division is done first, following the sign rules; the addition comes last.

• PRACTISE 1.2

- 1 Work out: a. $(-5)^2$ b. $(-9) \times (-4)$ c. $56 \div (-7)$.
- 2 Work out $(-6) \times (-2) - (-20) \div 4$.
- 3 Work out $4 \times (3 - 9)$ in two ways and check they agree.
- 4 Find the missing number: $(-36) \div \square = 9$.
- 5 a. Work out $(-2)^3$ b. Explain why a negative number cubed is negative.

UNIT 1 • TOPIC 1.3

Square roots and cube roots

Learn

A **square root** undoes squaring: $\sqrt{169} = 13$ because $13^2 = 169$. A **cube root** undoes cubing: $\sqrt[3]{343} = 7$ because $7^3 = 343$. Most numbers are not perfect powers, but you can trap their roots between two whole numbers.

■ WORKED EXAMPLE 1.3

Between which two consecutive whole numbers does $\sqrt{200}$ lie?

$$14^2 = 196 \text{ and } 15^2 = 225 \\ 196 < 200 < 225, \text{ so } 14 < \sqrt{200} < 15$$

Why: Because 200 sits between two consecutive squares, its root sits between the corresponding roots.

• PRACTISE 1.3

- 1 Work out: a. $\sqrt{196}$ b. $\sqrt[3]{1000}$ c. $\sqrt{81}$.
- 2 Between which two consecutive whole numbers does $\sqrt{90}$ lie?
- 3 Work out $\sqrt{144} - \sqrt{64}$.
- 4 I cube a number and get 512. What was the number?
- 5 Estimate $\sqrt{150}$ to one decimal place, then check with a calculator.

Indices

Learn

An **index** (plural indices) tells you how many times to multiply a number by itself: 3^4 means $3 \times 3 \times 3 \times 3 = 81$. The 4 is the index, and 3 is the base. Indices have three beautiful laws that make hard calculations easy.

- To **multiply** powers of the same base, **add** the indices: $3^4 \times 3^2 = 3^6$.
- To **divide** powers of the same base, **subtract** the indices: $5^7 \div 5^3 = 5^4$.
- Any number to the power **zero** is **1**: $7^0 = 1$.

■ WORKED EXAMPLE 1.4

Work out: a. $2^4 \times 2^3$ b. $(3^2)^3$ c. $10^5 \div 10^2$

$$\begin{aligned} \text{a. } 2^4 \times 2^3 &= 2^7 = 128 \\ \text{b. } (3^2)^3 &= 3^6 = 729 \\ \text{c. } 10^5 \div 10^2 &= 10^3 = 1000 \end{aligned}$$

Why: Adding, multiplying and subtracting indices turns long multiplications into one-step index operations.

• PRACTISE 1.4

- 1 Write in index form: a. $5 \times 5 \times 5$ b. $7 \times 7 \times 7 \times 7 \times 7$.
- 2 Work out: a. 2^6 b. 10^4 c. $(-2)^4$.
- 3 Work out, giving your answer as a power: a. $4^3 \times 4^2$ b. $6^8 \div 6^5$.
- 4 Work out: a. 9^0 b. $(5^2)^3$.
- 5 Work out 2^{10} . Is it bigger or smaller than 10^3 ?

∴ THINK LIKE A MATHEMATICIAN

The number 64 is special. It is 8^2 , it is 4^3 , and it is 2^6 . So 64 is a perfect square, a perfect cube and a perfect sixth power, all at once. Can you find another number less than 1000 that is both a perfect square and a perfect cube? Investigate and explain your method.

■ KEYWORDS

- **index:** the small raised number showing a power, as in 3^4 .
- **base:** the number being multiplied by itself in a power.
- **square root:** the number that squares to give the value under the root.
- **cube root:** the number that cubes to give the value under the root.

UNIT 1 • FINAL PROJECT

The prime detective

★ FINAL PROJECT

The prime detective

A mysterious number has been found. It is less than 100, it is a perfect square, and when you add its digits you get 10. Use everything you have learned in Unit 1 to find it, and present your reasoning like a detective's case file.

What to hand in

- The number, with a full justification.
- Its prime factorisation, written as a product of primes.
- A second mystery number of your own invention, with three clues, for a classmate.

How it will be judged

Criterion	Weight	--- ---	Correct number and justification	40%	Correct prime factorisation	30%	Quality of reasoning	20%	Originality of the invented puzzle	10%
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UNIT 1 · CHECK YOUR PROGRESS

- 1 Write 72 as a product of prime factors.
- 2 List the prime numbers between 20 and 40.
- 3 Work out: a. $(-6) \times (-7)$ b. $(-48) \div (-8)$.
- 4 Work out $(-4) \times (-3) + (-36) \div 9$.
- 5 a. Work out $\sqrt{225}$ b. Between which two whole numbers does $\sqrt{130}$ lie?
- 6 Write $5 \times 5 \times 5 \times 5$ in index form, and work out its value.
- 7 Work out, giving your answer as a power: a. $3^5 \times 3^2$ b. $7^9 \div 7^6$.
- 8 Work out: a. 8^0 b. $(2^3)^2$ c. $(-3)^3$.

UNIT 1 · EVALUATION

Be honest. For each outcome, colour the circle that matches how you feel: green means confident, amber means nearly there, red means needs more work.

I CAN ...	GREEN	AMBER	RED
find factors, multiples and primes	☐	☐	☐
write a number as a product of prime factors	☐	☐	☐
multiply and divide integers in several steps	☐	☐	☐
work with roots and estimate their values	☐	☐	☐
use index notation and the laws of indices	☐	☐	☐

UNIT 1 · WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- | | |
|--|--|
| ☐ Find factors, multiples and prime numbers. | ☐ Evaluate roots and trap them between whole numbers. |
| ☐ Write a number as a product of prime factors. | ☐ Use the three laws of indices. |
| ☐ Apply sign rules in multi-step calculations. | |

■ UNIT 1 ROUND-UP

factor · multiple · prime number · prime factor · index · base · square root · cube root

GLOSSARY

Unit 1

The mathematical words this unit has taught you, gathered in one place.

common factor: a number that is a factor of two or more numbers.

common multiple: a number that is a multiple of two or more numbers.

consecutive: following one after another in order.

cube number: a number made by multiplying an integer by itself twice, such as $27 = 3^3$.

cube root: the number that cubes to give the value under the root, as in $\sqrt[3]{27} = 3$.

digit: any of the symbols 0 to 9 used to write numbers.

divisible: able to be divided exactly, with no remainder.

factor: a number that divides another exactly.

highest common factor (HCF): the largest common factor of two or more numbers.

index: the small raised number showing a power, as in 5^2 .

integer: a positive or negative whole number, or zero.

inverse: the number that undoes another; the inverse of 4 is -4 .

inverse operation: an operation that reverses another; subtraction is the inverse operation of addition.

lowest common multiple (LCM): the smallest common multiple of two or more numbers.

multiple: the result of multiplying a number by a positive integer.

number line: a line on which numbers are placed in order.

product: the result of a multiplication.

quotient: the result of a division.

square number: a number made by multiplying an integer by itself, such as $36 = 6^2$.

square root: the number that squares to give the value under the root, as in $\sqrt{36} = 6$.

test for divisibility: a quick rule that tells you whether one number divides another without doing the division.

SOURCES & REFERENCES

Where everything came from

Every worked-example answer and exercise value in this book was checked by a computer before publication.

Factual notes can be checked here. Links were live as of July 2026.

§ UNIT 1 • INTEGERS

- 1 **Prime numbers and prime factorisation.** The Fundamental Theorem of Arithmetic: every whole number greater than 1 has a unique prime factorisation. <https://en.wikipedia.org/wiki/Fundamental_theorem_of_arithmetic>
- 2 **The laws of indices.** The rules for multiplying, dividing and raising powers, and the value of a number to the power zero. <<https://en.wikipedia.org/wiki/Exponentiation>>
- 3 **Sardinha festivals.** The Portuguese sardine-festival context used in the Unit 1 scenario is drawn from real summer festivals in Portugal.

PRIME BOOKS

Mathematics

Student Manual

very rule is

INSIDE THIS BOOK

- Worked examples before every exercise
- Fluency practice and problem-solving
- Calculator and non-calculator skills
- Cumulative unit reviews
- Exam-style questions with answers

Prime Books • Mathematics

Ages 12–13 • Lower Secondary

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